

SOF IMO — Topic Deep-Dive: Fractions

For Ambar, Class III CBSE

1. What a fraction is

A fraction is a part of a whole that has been cut into **EQUAL** parts. The word “equal” is what examiners test — 2 unequal pieces are NOT halves.

Cut into	Each part is called	Written as
2 equal parts	one-half	$\frac{1}{2}$
3 equal parts	one-third	$\frac{1}{3}$
4 equal parts	one-quarter (one-fourth)	$\frac{1}{4}$

Reading a fraction

In $\frac{3}{4}$: - 3 on top = **numerator** = how many parts we are talking about - 4 below = **denominator** = how many equal parts in total

Memory hook: **D**enominator is **D**own.

Wholes

- 2 halves make 1 whole. 3 thirds make 1 whole. 4 quarters make 1 whole.
- So $\frac{2}{2} = \frac{3}{3} = \frac{4}{4} = 1$ whole. And 2 quarters make a half.

2. Fraction of a collection

Rule: “ $\frac{1}{n}$ of a number” means **DIVIDE** the number by n .

Question	Working	Answer
$\frac{1}{2}$ of 16	$16 \div 2$	8
$\frac{1}{3}$ of 12	$12 \div 3$	4
$\frac{1}{4}$ of 20	$20 \div 4$	5
$\frac{1}{5}$ of 15	$15 \div 5$	3

Stretch rule: for $\frac{2}{3}$ of 12 — divide by the bottom, multiply by the top: $12 \div 3 = 4$, then $4 \times 2 = 8$.

Worked example: $\frac{3}{4}$ of 16 $\rightarrow 16 \div 4 = 4 \rightarrow 4 \times 3 = 12$.

3. Comparing fractions

For fractions with 1 on top (unit fractions): **The BIGGER the bottom number, the SMALLER the piece.**

Think of a roti: shared between 2 people you get a big piece ($\frac{1}{2}$); shared among 8 people you get a small piece ($\frac{1}{8}$).

- Order from smallest: $\frac{1}{8} < \frac{1}{4} < \frac{1}{3} < \frac{1}{2}$
- “Which is greater, $\frac{1}{3}$ or $\frac{1}{5}$?” $\rightarrow \frac{1}{3}$ (fewer people sharing, bigger piece)

4. “What fraction is shaded / left?” questions

- **Shaded:** parts shaded / total parts. 3 shaded out of 4 $\rightarrow \frac{3}{4}$.
- **Left / not shaded:** total – used, over total. Pizza of 8 slices, 3 eaten $\rightarrow 8 - 3 = 5$ left $\rightarrow \frac{5}{8}$ left.
- **Collections:** 4 ripe mangoes out of 12 $\rightarrow \frac{4}{12}$ of the mangoes are ripe.
- **Words:** 2 cricket days out of 7 week days $\rightarrow \frac{2}{7}$ of the week.

5. Common traps

Trap	Wrong thinking	Right thinking
Bigger denominator = bigger fraction	$1/8 > 1/2$	$1/8 < 1/2$ (more pieces = smaller piece)
Answering "eaten" when asked "left"	ate 3 of 8 $\rightarrow$ $3/8$ left	$5/8$ left
Flipping the fraction	3 of 4 parts = $4/3$	$3/4$
" $1/2$ of 16" = $16 - 2$	14	$16 \div 2 = 8$
Numerator/denominator mix-up	in $2/5$ denominator is 2	denominator is 5 (down)
Unequal parts	2 unequal parts = halves	halves must be EQUAL

6. High-frequency IMO question types

1. Name the fraction: "each part of a roti cut in 2 equal parts is ____".
2. Fraction of a number: " $1/2$ of 16 = ____".
3. Shaded/left fraction from a described figure or pizza story.
4. Compare unit fractions; order smallest to greatest.
5. Numerator/denominator identification.
6. Achiever's: two fractions of one collection (" $1/2$ are mangoes, $1/3$ are bananas, rest are apples").
7. Fraction with money: "spent $1/4$ of ₹40, how much left?"

7. Key facts checklist

- ☐ Halves (2), thirds (3), quarters (4) — parts must be equal
- ☐ Numerator on top, denominator down
- ☐ $1/n$ of a number = $\text{number} \div n$
- ☐ $2/3$ of a number = $(\text{number} \div 3) \times 2$
- ☐ Bigger denominator $\rightarrow$ smaller unit fraction
- ☐ $2/2$, $3/3$, $4/4$ all equal 1 whole; 2 quarters = 1 half
- ☐ "Left" = total minus used, over total

8. Quick drill (10 questions)

1. $1/2$ of 18 = ?
2. $1/4$ of 16 = ?
3. $1/3$ of 15 = ?
4. Which is smaller: $1/4$ or $1/6$?
5. A cake has 6 equal slices, 1 is eaten. What fraction is left?
6. How many quarters make one whole?
7. In $4/7$, what is the denominator?
8. $2/3$ of 9 = ?
9. Out of 20 marbles, 10 are blue. What fraction is blue?
10. Ambar spent $1/2$ of ₹30. How much is left?

Answers: 1) 9 2) 4 3) 5 4) $1/6$ 5) $5/6$ 6) $4/7$ 7) 7 8) $6 (9 \div 3 = 3, 3 \times 2 = 6)$ 9) $10/20 = 1/2$ 10) ₹15